AI Project: Deliverable 1 Guide

Problem Definition & Data Foundations

Deadline: 18 March, 2024

Overview

This first deliverable focuses on establishing the core ”Why” and ”What” of your project. You must transition from an approved idea to a concrete technical plan and a functional data pipeline. This involves Phase 1 (Problem Definition) and Phase 2 (Data Acquisition) of the project workflow.

1 Phase 1: Problem Definition

Students must document the theoretical and logical basis of their project according to their selected track.

1.1 Tracks A & B (Application Dev & Original R&D)

- **Problem Statement:** Provide a detailed explanation of the real-world pain point being addressed.
- **Target Audience:** Identify specifically who will use this tool or which research gap this project intends to fill.
- **AI Justification:** A technical argument explaining why Machine Learning/Deep Learning is necessary for this specific problem rather than traditional rule-based programming.

2 Phase 2: Data Acquisition & Preprocessing

A ”Data-Centric” approach is mandatory. You must demonstrate the movement from raw information to structured data.

- **Collection:** Provide evidence of data gathering (scripts for web scraping, API logs, or documentation of dataset downloads from Kaggle/UCI)
- **Cleaning Implementation:** Provide Python scripts that handle missing values, remove duplicates, and manage outliers.
- **Feature Strategy:** Document your implementation of normalization, scaling, and encoding of variables. Include initial exploratory data analysis (EDA) charts in your report.

3 Submission Checklist

To receive full marks, your submission must include the following components:

3.1 GitHub Repository

- A professional README.md file describing the project.
- A .gitignore file (do not upload large raw datasets to GitHub).
- A folder named /data scripts containing all Phase 2 code.
- **Note:** Commit history must show active participation from all group members.

3.2 Technical Report (LaTeX)

- A PDF report compiled from LaTeX covering all Phase 1 and Phase 2 requirements.
- Include visualizations of your raw data (e.g., class distributions, feature correlations).

4 Important Policies

- **Late Submissions:** Absolutely no late submissions will be accepted. Ensure you submit to Google Classroom before 11:59 PM on 23rd october.
- **Queries:** Reach out via official email for any technical clarifications.